Faster tests, greener cars

How machine learning can help plan the end-of-line test bench

Mohammad Bajbouj · ML project (Kaggle: Mercedes-Benz Greener Manufacturing) · neuefische AI PM bootcamp · October 2026

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Business case, approach and expected outcome

Main

Business case

- Every Mercedes-Benz car is tested on a test bench before delivery
- Test times differ from car to car: 72 to 265 seconds
- So the test bench is hard to plan – waiting and overrun cost time and energy

Approach

- Public Mercedes-Benz data: 4,209 tested cars, 376 columns
- Test our hypothesis with identical cars: same configuration = same test time?
- Predict the test time with three machine-learning methods, keep the simplest

Expected outcome

- A prediction of each car's test time from its configuration
- A clear answer to our hypothesis: how much of the test time can be predicted at all
- A proposal for planning the test bench, with buffer and next steps

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The challenge

Main

Unique

almost every car is configured differently

Every car

is tested on a test bench before delivery

72–265 s

test times differ from car to car – hard to plan the bench

Meaning: if we knew each car's test time in advance, we could plan the test bench better – less waiting, less energy.

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Our hypothesis

Main

Hypothesis (H2): Does the same car configuration always need the same test time?

1. Why it matters: if not, part of the test time can never be predicted from the configuration
2. Our approach: compare cars with identical configurations, then predict with three different machine-learning methods
3. Where we answer the hypothesis: slide 6 (identical cars) and slide 8 (three methods)

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What we worked with

Main

4,209

tested Mercedes-Benz cars (public Kaggle data)

376

columns per car – Mercedes-Benz hid the real names of the 376 columns

Seconds

one measured test time per car – what we predict

Meaning: real factory data, but anonymised: we see the values, not what each column means (→ slide 15).

→ explained on slide 15

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Finding 1: same car, different time

Main

Cars with exactly the same configuration still need different test times – by about 8 seconds.

≈ 60 %

of the differences in test time between different cars can be predicted, using the configuration as indicator – at most

≈ 40 %

comes from the test itself (e.g. human error) – no model can predict it

Meaning: the answer to our hypothesis is no – and this sets a clear limit for any prediction (→ slide 16).

→ explained on slides 16–17

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Finding 2: one column tells most of the story

Main

Each car is described by 376 columns. One column, X0, sorts the cars into a few typical test times.

57.5 %

of the differences in test time between cars are explained by the column X0 alone

375

the other 375 columns mostly repeat what X0 already says

Meaning: the test time depends mainly on the value in one column, X0 (→ slide 18).

→ explained on slide 18

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Our prediction

Main

Without a model

± 12.4 s

typical planning error if every car gets the same average time

With our model

± 8.0 s

about 4.5 s closer per car – already at the limit set by the test itself

Meaning: the average test time is about 100.6 s. Without a model the plan is off by about 12.4 s per car, with our model by about 8 s. Three different methods all landed here, so we keep the simplest (→ slide 19).

→ explained on slide 19

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How good is it?

Main

91 of 100

cars predicted within 10 seconds (tested on cars the model had not seen)

97 %

of the Kaggle winner's score (score = share of the differences explained), with one column (X0)

A few

very long tests are the big misses – they come from the test itself

Meaning: accurate enough to plan a test bench – not to promise each single car's time (→ slide 20).

→ explained on slide 20

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What it means for planning

Main

Car order

its configuration is known in advance

Prediction

expected test time for this car

Bench plan

a slot per car + a buffer for unusual tests

Expert view (Mohammad Bajbouj, automotive R&D): plan each car with its predicted test time to reduce energy use in the end, and keep a buffer for extraordinary test times.

The test itself does not get shorter – the plan gets more accurate: less waiting and less overrun on the bench.

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Why it matters at factory scale

Main

≈ 1,370

cars a day in the Mercedes-Benz plant Sindelfingen (315,069 cars in 2019, Mercedes-Benz AG)

4.5 s

more accurate plan per car

≈ 1.7 h

of more accurate bench planning every day – not shorter tests

Meaning: small per car, large per day – the energy saving is a hypothesis we propose to test in a pilot (→ slide 21).

→ explained on slide 21

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Next steps

Main

Pilot

one test bench, one day, then one month: measure idle time and energy before and after planning with predictions

Record more

long tests cannot be predicted from the configuration – record bench, tester and day to explain them (→ slide 22)

Know the limits

about 60 % is the limit; plan with predictions, don't promise single cars

→ explained on slide 22

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Key takeaways

Main

1. Even identical cars need different test times – at most about 60 % of the differences in test time between cars can be predicted
2. One column, X0, explains almost all of that predictable part
3. Our model plans each car about 4.5 seconds more accurately – 91 of 100 cars within 10 seconds
4. At factory scale this means about 1.7 hours of better planning per day – a pilot can show the energy saving

Meaning: the data, not the choice of method, sets the limit – and we already reach it.

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Explanation slides

For questions. Each slide names the main slide it belongs to.

Where the data comes from

Explain

1. Public Kaggle competition by Mercedes-Benz ("Greener Manufacturing", 2017): 4,209 tested cars with their measured test time
2. 376 columns per car: 8 text columns (e.g. X0) and 368 yes/no columns – the real names of the columns are hidden
3. Test times from 72.11 s to 265.32 s; one 265 s test looked faulty and was left out of training
4. A second set of 4,209 cars without test times is used by Kaggle to check every model fairly

→ belongs to main slide 5

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How we found identical cars

Explain

1. We compared all 376 columns of every car: 515 of 4,209 cars have at least one identical twin (217 groups)
2. Example: 9 identical cars needed between 106 and 128 seconds – up to 22 seconds apart
3. Across all groups, identical cars differ by about 8 seconds – this part is noise from the test itself
4. A person can check every group by hand; the twins behave like all other cars

→ belongs to main slide 6

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Where the 60 % comes from

Explain

1. All cars together: test times spread about 12.7 seconds around the average of 100.7 seconds
2. Identical cars: still about 8.1 seconds apart – the configuration is the same, so this cannot come from it
3. Comparing the two spreads: about 40 % of the differences is noise, so at most about 60 % can be predicted
4. The calculation step by step is in the engineering deck (slides 23–24)

→ belongs to main slide 6

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What X0 is

Explain

1. X0 is not a method – it is one column of the data; every car has one of 47 X0 values
2. Cars with the same X0 value have similar test times (e.g. groups around 75, 90, 100 or 108 seconds)
3. We cleaned the data: 66 columns never change or copy another column → 310 columns left
4. Adding these 310 columns did not improve the prediction – the 310 columns repeat what X0 already says

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Three methods, one result

Explain

1. Ridge: one straight-line formula – it cannot see combinations of columns. Random Forest: many decision trees, averaged – it can catch combinations Ridge misses. XGBoost: trees that fix each other's errors – usually the strongest for data like ours
2. Fair test: each method learned from part of the cars and was checked on cars it had not seen, in 5 rounds
3. All three: about 8 seconds typical error, about 59 % of the differences explained – equal
4. The stronger methods found nothing extra – so the limit is in the data, not in the method, and we keep the simplest: Ridge with X0

→ belongs to main slide 8

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What the scores mean

Explain

1. "Explained" = how much of the differences between cars the model can account for – not an error rate

2. Kaggle check on 4,209 hidden cars: our model 54.1 %, competition winner 55.6 %, limit about 59.7 %

3. So we reach 97 % of the winner's score and 91 % of what is possible – with one column (X0)

4. 91 of 100 cars are predicted within 10 seconds; the big misses are a few very long tests

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The factory-scale estimate

Explain

1. Production: the Mercedes-Benz plant Sindelfingen built 315,069 cars in 2019 (Mercedes-Benz AG, Umwelterklärung 2024, p. 35) – the last normal year before COVID, closest to the Kaggle data (2017)
2. Per day: about 1,370 cars, assuming about 230 working days (our assumption)
3. 4.5 seconds more accurate per car × 1,370 cars ≈ 6,165 seconds ≈ 1.7 hours of better planning per day
4. Energy saving is a hypothesis: the data has no energy values, and we don't know which plant the data comes from

→ belongs to main slide 11

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Why long tests cannot be predicted

Explain

1. 50 of 4,209 tests took unusually long (over 136 seconds)
2. Identical twins of these cars ran normally – in all 9 cases
3. The best warning rule is right for only 1 in 39 flagged cars – about twice as good as chance
4. So the extra time comes from the test run (bench, tester, day) → future work: record these data and test a targeted buffer

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